

CARDIAC IMAGING

Quantifying myocardial oxygen consumption and efficiency with motion-resolved cardiac MRI

Li-Ting Huang^{1,2†}, Chia-Chi Yang^{1†}, Guan Wang^{3†}, Henghui Zhang³, Ranran Zhang³, Hao Ho⁴, Archana Malagi¹, Yuheng Huang^{1,5,6,7}, Xinqi Li¹, Ghazal Yoosefian¹, Xinheng Zhang¹, Ziyang Long¹, Xiaoming Bi⁸, Janet Wei⁹, Alan C. Kwan¹⁰, Michael D. Nelson¹¹, C. Noel Bairey Merz⁹, Daniel Berman¹², Anthony Christodoulou^{7,13}, Debiao Li¹, Rohan Dharmakumar^{5,6}, Hsin-Jung Yang^{1*}

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Relentless mechanical work of the heart is powered by continuous oxygen consumption. How the heart uses oxygen is a defining feature of its health. Invasive studies have established that impaired oxygen consumption by the myocardium predicts contractile dysfunction and adverse outcomes. Despite its importance, noninvasive quantification of myocardial oxygen use remains limited. Magnetic resonance imaging (MRI) signal is known to be sensitive to blood oxygenation and has the potential to quantify myocardial oxygen consumption noninvasively, without exogenous contrast agents and free of ionizing radiation. However, its clinical translation has been impeded by the need for complex biophysical calibration, vulnerability to imaging artifacts and consistent vital motions, and the requirement of lengthy acquisition times. Here, we introduce a rapid, self-calibrated cardiac MRI framework that overcomes these barriers through high-resolution, motion-resolved coronary sinus oximetry, which can quantify myocardial oxygen extraction of the whole heart within 3 minutes. We optimized the imaging parameters via numerical simulations and validated them against invasive coronary sinus catheterization in a porcine model. We combined the method with clinical MRI sequences and demonstrated the feasibility of quantifying myocardial oxygen consumption and myocardial oxygen efficiency in patients with and without heart failure secondary to myocardial infarction in a single institution. This needle-free approach establishes a practical framework for noninvasive characterization of myocardial oxygen metabolism. It holds the potential to facilitate early disease detection, inform personalized therapeutic strategies, and guide the development of cardiometabolic therapies aimed at addressing the ongoing heart failure epidemic.

INTRODUCTION

Oxygen metabolism is central to all human organs. Particularly, the heart is highly energy dependent, requiring a continuous supply of oxygen to sustain its function (1). Impairment in myocardial oxygen supply, demand, or its utilization efficiency can cause mechanical dysfunction, which ultimately leads to heart failure (HF). Recognized as a growing global health crisis, HF is the common downstream manifestation of diverse cardiac and systemic diseases. Among the various etiologies, coronary artery disease is the leading cause of HF and was reported to

be associated with a higher risk of mortality in patients with HF. In coronary artery disease, narrowing or obstructing coronary arteries reduces myocardial perfusion, creating a mismatch between oxygen supply and demand. The mismatch initiates myocardial ischemia, disrupts oxygen metabolism, accelerates the myocardial structural and functional deterioration, and results in HF. Direct measurement of myocardial oxygen consumption (MVO_2) and utilization efficiency holds transformative potential for early diagnosis, risk stratification, and enabling precision cardiometabolic therapy for the HF epidemic (2–4). Impaired oxygen efficiency, even in the presence of preserved global function, may reflect early metabolic dysfunction and signal an elevated risk of irreversible myocardial remodeling and progression to HF (5, 6).

Despite its clinical importance, the accessibility of measuring MVO_2 directly in today's clinical practice is limited. Indirect markers of myocardial metabolism, such as clinical symptoms (7), wall motion abnormalities (8), electrocardiographic changes (9), or circulating biomarkers (10), are often adopted to indicate impaired cardiac metabolism. However, they are not specific to myocardial energetics and reflect late-stage symptoms instead of the cause of the disease, making them unable to guide early interventions.

MVO_2 involves assessment of the difference in arteriovenous oxygen content, where invasive catheterization for measuring arterial and venous blood in the coronary circulation is required. The gold standard to measure blood oxygen saturation in the coronary sinus is coronary sinus catheterization. Although accurate, the approach is invasive, associated with procedural risk, and unsuitable for early-stage disease assessment or repeat monitoring (11). Noninvasive approaches such as positron emission tomography (PET with ^{11}C -acetate

¹Biomedical Imaging Research Institute, Cedars-Sinai Medical Center, Los Angeles, CA 90048, USA. ²Department of Medical Imaging, National Cheng Kung University Hospital, College of Medicine, National Cheng Kung University, Tainan 704, Taiwan. ³Department of Radiology, First Hospital of China Medical University, Shenyang 110001, Liaoning, China. ⁴Department of Statistics, University of California at Los Angeles, Los Angeles, CA 90095, USA. ⁵Cardiovascular Imaging Research Center, Medical Imaging Research Institute, Department of Radiology & Imaging Sciences, Indiana University School of Medicine, IN 46202, USA. ⁶Krannert Cardiovascular Research Center, Division of Cardiovascular Medicine, Department of Medicine, Indiana University School of Medicine, IN 47408, USA. ⁷Department of Bioengineering, University of California, Los Angeles, Los Angeles, CA 90095, USA. ⁸Siemens Medical Solutions USA Inc., Los Angeles, CA 90045, USA. ⁹Division of Cardiology, Women's Heart Center, Heart Institute, Cedars-Sinai Medical Center, Los Angeles, CA 90048, USA. ¹⁰Department of Cardiology, Smidt Heart Institute, Cedars-Sinai Medical Center, Los Angeles, CA 90048, USA. ¹¹Department of Kinesiology, University of Texas at Arlington, Arlington, TX 76019, USA. ¹²Division of Artificial Intelligence in Medicine, Departments of Medicine, Biomedical Sciences, and Imaging, Cedars-Sinai Medical Center, Los Angeles, CA 90048, USA. ¹³Department of Radiological Sciences, David Geffen School of Medicine at UCLA, Los Angeles, CA 90095, USA.

*Corresponding author. Email: hsin-jung.yang@cshs.org

†These authors contributed equally to this work.

and ^{15}O -water) have been proposed to measure MVO_2 through the tricarboxylic acid cycle (12, 13). However, the need for radiotracer injection, exposure to ionizing radiation, and reliance on an on-site cyclotron limit its accessibility and clinical scalability (14).

Magnetic resonance imaging (MRI) offers a radiation and contrast-free method with inherent sensitivity to blood oxygenation via the blood oxygen level-dependent (BOLD) signal. BOLD MRI has transformed functional neuroimaging by enabling in vivo mapping of brain activity through blood susceptibility changes without the use of exogenous contrast agents (15–17). Although attempted in the heart (18–24), accurate quantification of cardiac oxygen consumption and utilization of the whole heart remains challenging because of technical challenges, including the need for complex calibration of magnetic resonance (MR) blood oximetry models, low accuracy from insufficient BOLD contrast-to-noise ratio, motion sensitivity and limited spatial coverage in gated or breath-hold scans, and the absence of direct validation against invasive gold standards (21–23, 25, 26). Here, we present a rapid, self-calibrated, and motion-resolved cardiac MRI framework that enables free-breathing coronary sinus blood oximetry and quantifying whole-heart myocardial oxygen extraction without contrast agents or pharmacologic stress.

RESULTS

Myocardial oxygen extraction fraction quantification with high-resolution, motion-resolved, self-calibrated coronary sinus MR blood oximetry

The motion-resolved, quantitative cardiac MRI approach uses a continuous T_2 -inversion recovery (IR)-prepared three-dimensional (3D) stack-of-stars gradient echo sequence for coronary sinus (CS) coverage throughout the cardiac cycle (fig. S1A). Acquired k -space data were reconstructed into a 6D tensor, preserving spatial and temporal signal change. A low-rank tensor reconstruction framework was adopted to facilitate tensor completion and motion-resolved multiparametric mapping. To enable precise blood oxygen saturation ($\text{SbO}_2\%$) quantification, we used a self-calibration oximetry model leveraging the Luz-Meiboom exchange model and a two-compartment T_1 -weighted model to calibrate the biophysical parameters and correct for blood T_1 and T_2 deviation from the conventional native, single-compartment T_1 and T_2 relaxometry (27). Representative motion-resolved BOLD images (fig. S1B) confirmed that systolic imaging, with a minimal temporal footprint, enables visualization of CS lumen with high fidelity. The integration of image contrast from the T_2 -IR preparation pulses enabled the delineation of the blood pool from surrounding tissues and provided the necessary BOLD sensitivity for the determination of $\text{SbO}_2\%$.

Numerical simulations for parameter optimization and oximetry precision analysis

Numerical simulations were first conducted to optimize the preparation parameters for the proposed sequence following the self-calibrated oximetry model. BOLD signal sensitivity ($\Delta\text{BOLD signal}/\Delta\text{SbO}_2\%$) was calculated on the basis of the spin refocusing time (τ_{180}), blood hematocrit (Hct), and $\text{SbO}_2\%$ (Fig. 1A). A set of τ_{180} values (7.5, 15, 22.5, 27.5, and 30 ms) was selected to ensure that sufficient BOLD sensitivity ($\Delta\text{BOLD signal}/\Delta\text{SbO}_2\% > 0.8$ of the theoretical maximum) could be achieved throughout the targeted CS $\text{SbO}_2\%$ range (Fig. 1B). A noise propagation analysis was then conducted to investigate the oximetry accuracy of the selected parameters. An image with a

signal-to-noise ratio (SNR) between 7 and 100 was tested to cover the SNR range of acquired in vivo images, where the SNR is ~ 15 in the CS (Fig. 1C). The simulation showed a less than 2% standard deviation when measuring $\text{SbO}_2\%$ across the range of CS blood when adopting the in vivo SNR (Fig. 1D).

Confounder-corrected cardiac blood oxygenation and CS-based cardiac magnetic resonance quantification of whole-heart oxygen extraction fraction, myocardial blood flow, and MVO_2 comparable to the invasive reference

To validate the proposed method, we directly measured CS oxygen saturation via invasive catheter and CS blood flow by a thermodilution method and compared these “invasive gold standard” measures against the proposed cardiac MRI equivalents (phase contrast CS flow and BOLD-derived blood oxygen saturation, respectively) in $N = 10$ healthy female domestic farm pigs. In addition to measuring MVO_2 , established cardiac magnetic resonance (CMR) techniques were used to measure myocardial mass, volume, and cardiac work (28, 29). Integrating MVO_2 with cardiac morphology and hemodynamics, myocardial oxygen efficiency (MOE), defined as the ratio of kinetic energy associated with cardiac work to the chemical energy derived from MVO_2 , was derived (30). The schematic diagram illustrates the study design, showing measurements obtained at the baseline and after 3 min of continuous adenosine infusion (fig. S2).

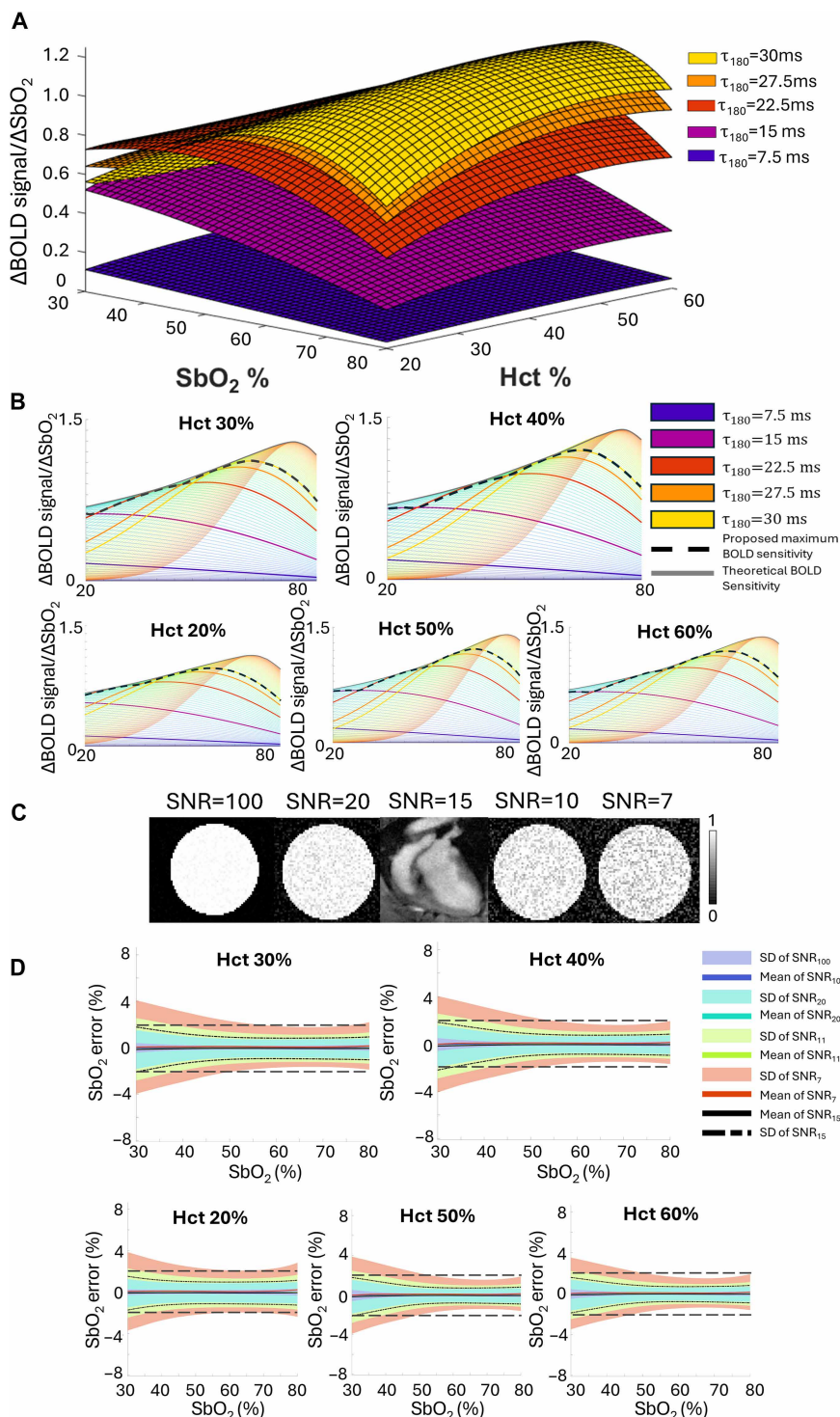
T_2 -IR images were acquired in two views (I: a midventricular short axis plane; II: a long axis plane covering a cross section of the CS) of the heart covering the left ventricle (LV), right ventricle (RV), and CS. Images were reconstructed using the low-rank tensor formulation with conventional naïve single-compartment T_1 and T_2 relaxometry and the proposed T_1 - T_2 -based oximetry model. $\text{SbO}_2\%$ measurements in the bilateral ventricles and CS were validated against the invasive ground truth. In the representative pig, the proposed model derived $\text{SbO}_2\%$ values in LV (98.22%) and CS (62.16%) that more closely matched invasive measurements (100 and 61.00%, respectively), whereas the naïve single-compartment relaxometry model showed substantial deviations (89.83 and 26.57%, respectively) (Fig. 2A).

The linear regression between CMR-based $\text{SbO}_2\%$ measurements and invasive $\text{SbO}_2\%$ was linearly correlated in both the naïve single-compartment relaxometry model [correlation coefficient (r) = 0.74, $P = 0.002$] and proposed oximetry model ($r = 0.99$, $P = 0.000$) (Fig. 2B). Bland-Altman plots revealed reduced bias with the proposed model (bias, -0.46% ; 95% limits of agreement, -24.31 to 23.39%) compared with the conventional relaxometry model (bias, -10.63% ; 95% limits of agreement, -42.50 to 21.24%) (Fig. 2C). In addition, the naïve model could not differentiate SbO_2 between LV and RV (84.70% versus 71.79%, $P = 0.725$), which were distinctly separated from the proposed method with its improved accuracy (LV versus RV = 96.23% versus 79.94%, $P = 0.020$; RV versus CS = 79.94% versus 57.24%, $P = 0.003$) (Fig. 2D). The results demonstrate the ability of the proposed model to substantially decrease the $\text{SbO}_2\%$ bias, enhance the goodness of fit, and accurately quantify the oxygen extraction fraction (OEF) in the heart.

To test the capability of the proposed CMR framework to measure OEF and MVO_2 , CMR-based parameters were measured in pigs and compared against the invasively measured ground truth. The linear regression analysis demonstrated a correlation between CMR-derived OEF and invasive OEF ($r = 0.95$, $P < 0.01$). Correlations were observed between CMR myocardial blood flow (MBF) and invasive

Fig. 1. Sensitivity and accuracy analysis of the proposed MR oximetry sequence using numerical simulations.

(A) The BOLD sensitivity ($\Delta\text{BOLD signal}/\Delta\text{SbO}_2\%$) was plotted in a 3D view. The BOLD sensitivity at different combinations of $\text{SbO}_2\%$ and Hct was illustrated with a range of τ_{180} values (purple: 7.5 ms; magenta: 15 ms; red: 22.5 ms; orange: 27.5 ms; yellow: 30 ms). (B) Cross sections of the simulated BOLD sensitivity plot in the $\text{SbO}_2\%$ dimensions under different Hct. The opaque curves represent the BOLD sensitivity with a range of τ_{180} values (0 to 80 ms). The curves formed a BOLD sensitivity contour (gray line), representing the maximum BOLD signal changes in each $\text{SbO}_2\%$ and Hct. We identified five τ_{180} values [7.5, 15, 22.5, 27.5, and 30 ms; shown in solid lines with the same colors presented in (A)] to ensure that 80% of the maximum signal variation was achieved in the targeted CS $\text{SbO}_2\%$ and Hct range. (C) Simulated Gaussian noises were added to represented circles of the blood pool with SNR = 7 to 100 and illustrated along an in vivo CS image that was measured at SNR = 15 (middle). (D) Numerical simulation was conducted to fit the BOLD curves and derive $\text{SbO}_2\%$ under different SNRs. The results from 1000 repetitions were presented with means (solid lines) and standard deviations (color bands) using different SNRs (red: 7; yellow: 11; teal: 20; purple: 100). SNR = 15 was used to represent in vivo oximetry data (black lines). The plots demonstrated that the $\text{SbO}_2\%$ errors have a standard deviation (black dotted line) smaller than 2% across the target Hct and $\text{SbO}_2\%$ range at the in vivo SNR.



measurements ($r = 0.64$, $P = 0.01$), as well as between CMR MVO_2 and invasive MVO_2 ($r = 0.70$, $P < 0.01$) (Fig. 2E). Bland-Altman analyses demonstrated favorable agreement between the CMR and the invasive ground truth across all parameters (OEF: bias, -0.01 ; 95% limits of agreement, -0.11 to 0.10 ; MBF: bias, 10.27 ml/min; 95% limits of agreement, -127.77 to 148.31 ; MVO_2 : bias, 0.50 ml/min; 95% limits of agreement, -4.40 to 5.39 ; Fig. 2F).

Motion-resolved CMR pipeline for measuring OEF, MVO_2 , and MOE across physiological states

To test the proposed CMR framework for characterizing myocardial oxygen metabolic changes under different physiological conditions, OEF, MVO_2 , and MOE were assessed under resting baseline conditions and then during adenosine administration in healthy pigs. Representative images demonstrated that CS SbO_2 and blood flow rates were higher during adenosine infusion compared with the baseline, whereas the LV stroke volume (calculated as the difference between end-diastolic and end-systolic volumes) remained comparable between conditions (Fig. 3, A to C). The global MBF, OEF, global MVO_2 , and MOE measured by CMR showed results comparable to those from the invasive measurements (all $P > 0.05$; Table 1). In the comparative analysis of CMR parameters between the baseline and adenosine status, MBF was significantly increased with adenosine (from 176.39 ± 58.58 to 237.02 ± 72.77 ml/min, $P = 0.013$), resulting in

a reduction in CS OEF (from 0.46 ± 0.10 to 0.22 ± 0.05 , $P < 0.001$; Fig. 3D). As such, MVO_2 was reduced with adenosine (from 8.12 ± 2.81 to 4.97 ± 1.81 , $P = 0.031$), consistent with a reduction in rate pressure product (RPP) (from 5725.36 ± 1019.16 to 4055.93 ± 924.55 , $P = 0.027$) (Fig. 3D) (31). The adenosine response in anesthetized swine differed from that in conscious humans. In conscious humans, adenosine lowers blood pressure and triggers a reflex

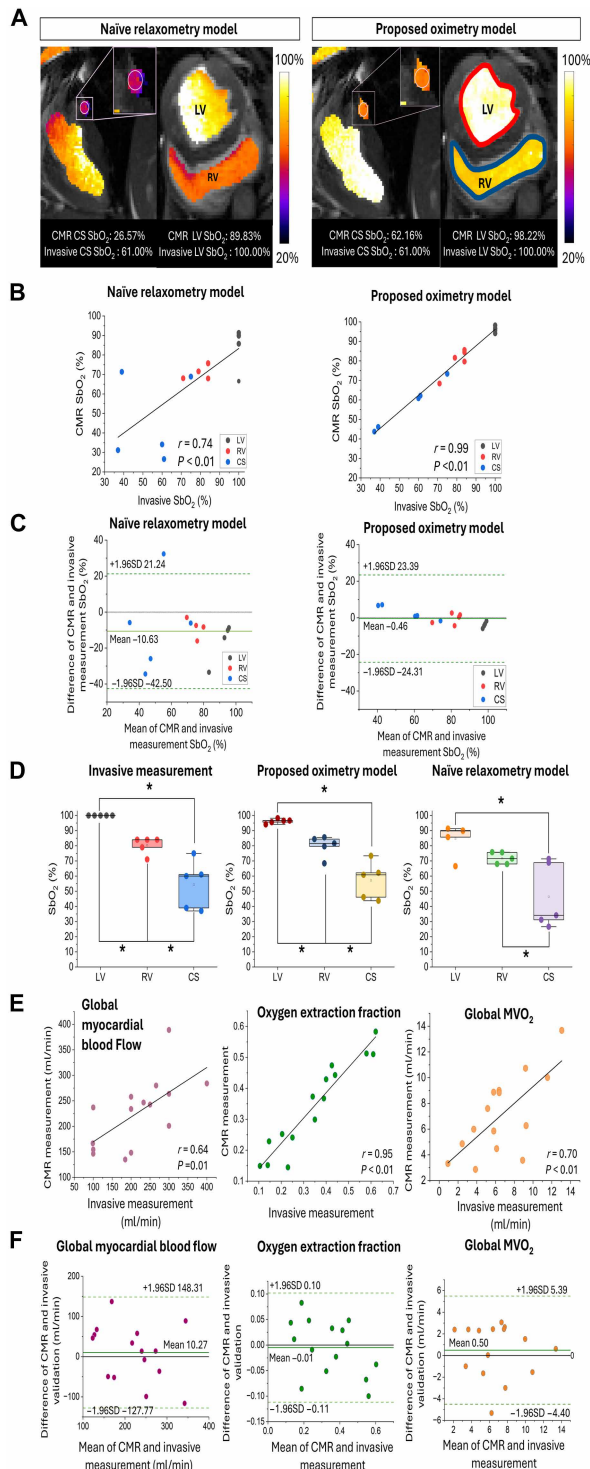


Fig. 2. The proposed MR oximetry model can correct for the bias from the naïve single-compartment relaxometry model and derive accurate $\text{SbO}_2\%$, OEF, and MVO_2 compared to the invasive ground truth in pigs under the baseline and adenosine infusion. (A) Representative images of the $\text{SbO}_2\%$ maps derived from the naïve relaxometry model and proposed oximetry models are presented. Color masks in the CS, LV, and RV represent SbO_2 . SbO_2 measurements were as follows: CS: 26.57% and LV: 89.83% by the naïve relaxometry model; CS: 62.16% and LV: 98.22% by the proposed oximetry model; CS: 61.00% and LV: 100.00% by invasive catheterization. **(B)** The linear regression plot (left) shows the correlation between SbO_2 measured by the naïve relaxometry model and invasive catheterization ($r = 0.74$, $P < 0.01$) across the LV ($n = 5$), RV ($n = 5$), and CS ($n = 5$). The plot (right) shows the correlation between SbO_2 measured by the proposed oximetry model and invasive catheterization ($r = 0.99$, $P < 0.01$) across the LV ($n = 5$), RV ($n = 5$), and CS ($n = 5$). The solid line represents the linear regression fit. Correlation was assessed using Pearson correlation analysis. **(C)** The Bland-Altman plot shows the error of the proposed MR oximetry model compared with the conventional model. The right plot shows the mean difference in SbO_2 between the naïve relaxometry model and invasive catheterization of -10.63% (95% limits of agreement: $\pm 31.87\%$). The left plot shows the mean difference in SbO_2 between the proposed oximetry model and invasive catheterization of -0.46% (95% limits of agreement: $\pm 23.85\%$). Each plot includes $n = 5$ measurements from the CS, LV, and RV. The green solid line represents the means. The dashed black line is zero. The upper and lower dashed green lines represent the 95% limits of agreement. **(D)** The box plot (right) shows differences in SbO_2 among the LV, RV, and CS ($n = 5$ per chamber, all $P < 0.05$) measured by invasive catheterization. The box plot (middle) shows differences in SbO_2 among the LV, RV, and CS ($n = 5$ per chamber, all $P < 0.05$) measured by the proposed oximetry model. The box plot (left) shows differences in SbO_2 between LV and CS and between RV and CS ($n = 5$ per chamber, $P < 0.05$) measured by the naïve relaxometry model. The middle line in the boxes represents the median. The lower and upper bounds of the boxes represent the 25th and 75th percentiles, respectively. The whiskers represent the minimum and maximum values. Differences between groups were analyzed using a one-way repeated-measures ANOVA followed by Fisher's least significant difference post hoc test. $*P < 0.05$. **(E)** The linear regression plot (left) shows the correlation between global MBF measured by CMR and invasive catheterization ($n = 15$; $r = 0.64$, $P = 0.01$). The linear regression plot (middle) shows the correlation between OEF measured by CMR and invasive catheterization ($n = 15$; $r = 0.95$, $P < 0.01$). The linear regression plot (right) shows the correlation between OEF measured by CMR and invasive catheterization ($n = 15$; $r = 0.70$, $P < 0.01$). The solid line represents the linear regression fit. Correlation was assessed using Pearson correlation analysis. **(F)** Bland-Altman plots of global MBF, OEF, and global MVO_2 . The right plot shows that the mean difference in global MBF between CMR measurement and invasive catheterization was 10.27 ml/min (95% limits of agreement: ± 138.04 ml/min). The middle plot shows a mean difference in OEF between CMR measurement and invasive catheterization of -0.01 (95% limits of agreement: ± 0.11). The left plot shows a mean difference in global MVO_2 between CMR measurement and invasive catheterization of 0.50 ml/min (95% limits of agreement: ± 4.90 ml/min). $n = 15$ for all plots. The green solid line represents the means. The solid black line is zero. The upper and lower dashed green lines represent the 95% limits of agreement.

increase in heart rate via sympathetic activation (32). This response is blunted in anesthetized pigs, leading to a greater drop in blood pressure without a heart rate change (33). CMR-derived MOE was largely preserved between the baseline and adenosine (23.67 ± 14.19 versus 27.74 ± 14.70 , $P = 0.688$) (Fig. 3D), which is consistent with prior observations in healthy pig hearts (34). The details of invasive and CMR-derived measurements are presented in Table 1.

Detection of impaired oxygen consumption and efficiency across HF phenotypes using the proposed CS-based CMR oximetry

To translate this work clinically, we evaluated CS oxygen saturation, MBF, and MOE in 33 human participants, comprising 11 healthy controls [mean age: 63.82 ± 11.57 years; male: 10 (90.91%)] and 22 patients with HF (35) with a history of prior myocardial infarction

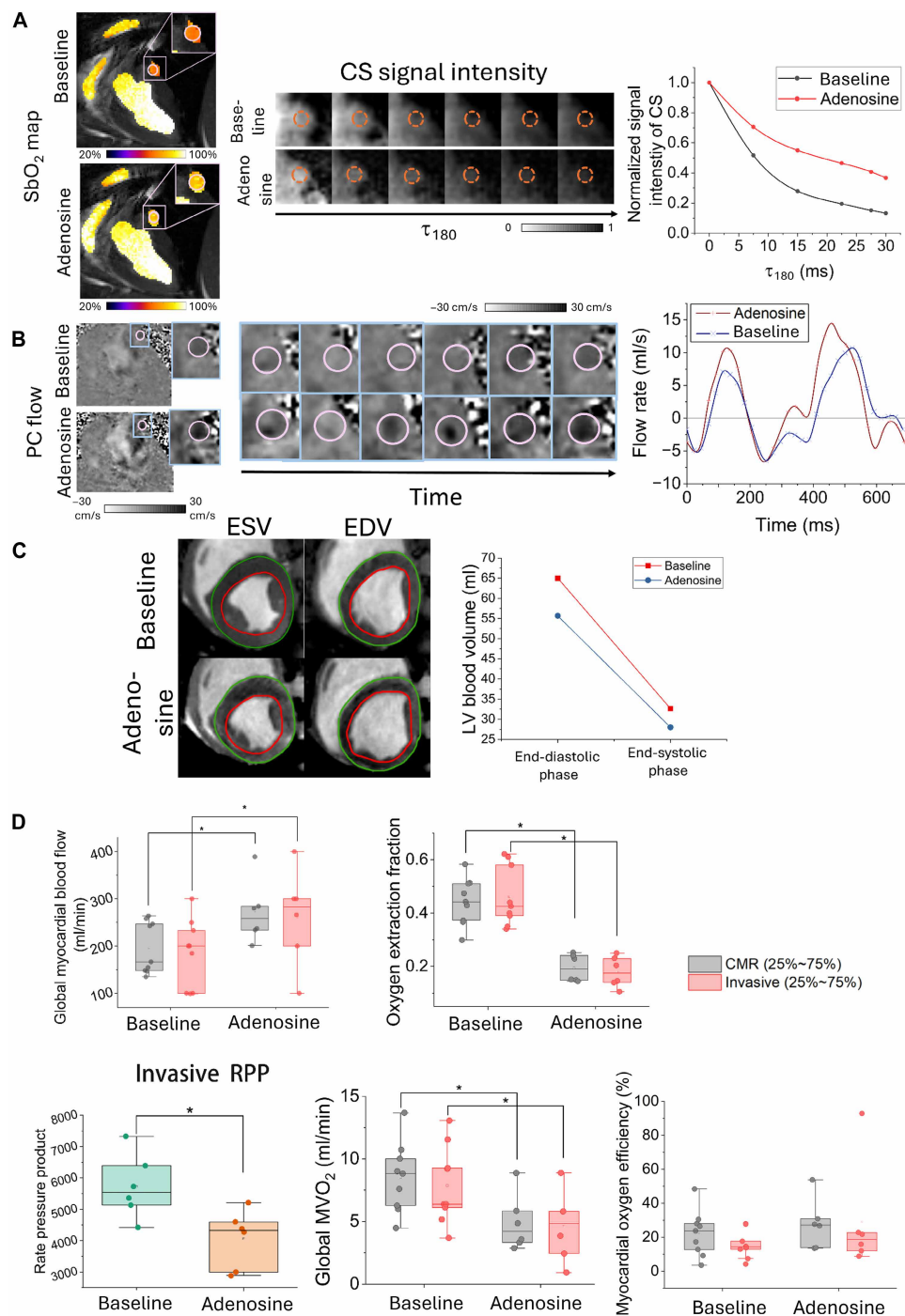


Fig. 3. Determinants of MVO_2 at the baseline and during adenosine administration. (A) Representative images of one pig show CS SbO_2 at the baseline and during adenosine infusion (left) with associated BOLD signal changes (middle). The color scale bar indicates SbO_2 : 20 to 100%. The grayscale bar indicates signal intensity: 0 or 1 (normalized). The plot (right) shows normalized CS signal intensity versus τ_{180} at the baseline and during adenosine infusion. (B) Representative 2D phase contrast MR images (left and middle) demonstrate CS blood flow patterns during the baseline and adenosine status. Quantitative flow analysis (right) shows CS flow rates at the baseline and during adenosine infusion. The grayscale bar indicates flow velocity: -30 to 30 cm/s. PC, phase contrast. (C) Representative images (left) of the ESV and EDV in the baseline and adenosine status. The plot (right) shows LV stroke volume at the baseline and during adenosine infusion, calculated as EDV minus ESV. (D) Box plots compare global MBF (top left) and OEF (top right) measured by CMR and invasive catheterization at the baseline ($n = 9$) and during adenosine infusion ($n = 6$). The plot (bottom left) compares the invasive RPP at the baseline and during adenosine infusion ($n = 6$). Box plots compare global MVO_2 (bottom middle) and MOE (bottom right) measured by CMR and invasive catheterization at the baseline ($n = 9$) and during adenosine infusion ($n = 6$). The middle lines in the boxes represent the medians. The lower and upper bounds of the boxes represent the 25th and 75th percentiles, respectively. The whiskers represent the minimum and maximum values. Differences between conditions were analyzed using a paired two-tailed t test. $*P < 0.05$.

Table 1. Comparison of OEF, MBF, MVO₂, and MOE between invasive and CMR-derived measurements at the baseline and during adenosine status. The data are presented as the means (standard deviation).

Parameters	Invasive			CMR			Invasive versus CMR	
	Baseline	Adenosine	P	Baseline	Adenosine	P	P	
	(n = 6*)	(n = 6*)		(n = 9)	(n = 9)		Baseline (n = 9)	Adenosine (n = 6*)
OEF	0.45 (0.11)	0.18 (0.05)	0.001	0.46 (0.10)	0.22 (0.05)	0.000	0.303	0.552
MBF (ml/min)	197.57 (74.77)	273.71 (99.01)	0.009	176.39 (58.58)	237.02 (72.77)	0.013	0.531	0.822
MVO ₂ (ml/min)	7.87 (3.47)	4.83 (2.64)	0.008	8.12 (2.81)	4.97 (1.81)	0.031	0.395	0.841
MOE (%)	18.05 (8.27)	28.99 (31.74)	0.688	23.67 (14.19)	27.74 (14.70)	0.688	0.508	1.000

*In a total of 10 individuals, 4 pigs did not survive long enough to receive adenosine administration in the invasive procedure or cardiac MRI acquisition.

(MI) caused by coronary artery disease [mean age: 60.45 ± 10.44 years; male: 19 (86.36%)] (Fig. 4A). The diseased participants demonstrated significantly altered cardiac structure and function compared with controls [ejection fraction (EF) = 34.07 ± 16.85%, LV mass index = 72.82 ± 21.13 g/m², LVESVI (left ventricular end-systolic volume) = 84.12 ± 41.76 ml/m², and LVEDVI (left ventricular end-diastolic volume index) = 121.98 ± 39.59 ml/m²; all *P* < 0.05]. Among which, on the basis of EF, 17 patients with reduced EF ≤ 40% were classified as HF with reduced EF (HFrEF), whereas 5 with preserved EF ≥ 50% were classified as HF with preserved EF (HFpEF) (35). In addition, the patients were classified for adverse LV structural remodeling using age- and sex-adjusted upper normal limits for LVEDVI (36). Seventeen (77.27%) patients were categorized as having left ventricular dilation (LVD+), and five (22.73%) were categorized as not having left ventricular dilation (LVD−). The proposed CS oximetry was successfully measured in all individuals regardless of their heart rhythm status. The average of R-R interval was not significantly different between the controls and patients who had MI (813.08 ± 118.343 versus 831.79 ± 132.10, *P* = 0.62), whereas the coefficient of variation of the R-R interval was significantly higher in the patients who had MI (0.04 ± 0.02 versus 0.05 ± 0.04, *P* < 0.05) (fig. S3). Four participants in the patient cohort were excluded from MBF-related analyses because of imaging artifacts from the prospectively gated breath-hold phase contrast image acquisition (e.g., excessive CS displacement from cardiac motion, gating failure, and severe oblique angulation of the CS throughout the cardiac cycle).

Visual comparison of CS SbO₂% maps and LV geometry images between a control individual and a patient with HF who had MI demonstrated reduced CS SbO₂% and impaired LV contractility from end-diastole to end-systole in the patient who had MI (Fig. 4B). OEF was significantly elevated in the HFpEF group compared with the HFrEF group (0.60 ± 0.04 versus 0.50 ± 0.08, *P* = 0.017) (Fig. 4C). Although the difference was not statistically significant, a trend emerged indicating that the HFrEF group had the highest global MVO₂ (8.16 ± 3.39 ml/min), followed by the HFpEF group (7.85 ± 2.20 ml/min) and the controls (6.67 ± 1.31 ml/min). The combined assessment of MOE and EF enabled the reclassification of patients with HFpEF but impaired MOE, which may facilitate the early detection of subclinical cardiac dysfunction (Fig. 4D). The stroke work reflects that the mechanical output of LV was significantly lower in the HFrEF group compared with the controls (45.73 ± 18.00 J versus 99.08 ± 17.31 J, *P* < 0.001

and the HFpEF group (45.73 ± 18.00 J versus 67.96 ± 17.64 J, *P* = 0.008). As a result, MOE was significantly lower in the HFrEF group compared with the controls (29.62 ± 8.63% versus 75.68 ± 13.11%, *P* < 0.001) and in the HFpEF group compared with the controls (43.92 ± 9.18% versus 75.68 ± 13.11%, *P* = <0.001) (Table 2). The scatterplots show the distribution of OEF, MBF, and global MVO₂ in the controls and patients who had MI against left ventricular EF (fig. S4). The patients with HF and LVD+ who had MI exhibited significantly reduced MOE compared with controls (28.38 ± 7.58% versus 75.68 ± 13.11%, *P* < 0.001) and patients with HF and LVD− who had MI (28.38 ± 7.58% versus 44.28 ± 7.99%, *P* = 0.016), indicating impaired myocardial energy efficiency associated with adverse postinfarction remodeling (fig. S5).

In addition to the patients with HF, we conducted a pilot study in humans to validate the reproducibility and technical robustness of the CS blood oximetry technique. Three patients were recruited to conduct right heart catheterization for clinical indications within 1 week of CMR. CS blood samples were collected during catheterization for oxygen saturation analysis. There was no significant difference in CS SbO₂ between invasive measurement and CMR assessment (41.90 ± 2.59% versus 43.48 ± 3.45%, *P* = 0.56).

DISCUSSION

In this study, we developed a comprehensive, noncontrast, radiation-free cardiac MRI framework that quantifies OEF, MVO₂, and MOE using standard MRI hardware. Tissue-based BOLD-sensitive MR imaging, T₂/T₂* relaxometry-based oximetry, and susceptibility mapping have provided proof-of-concept insights into regional ischemia and tissue viability (20, 21, 37–40), but these methods remain model-dependent and are strongly influenced by confounders, such as blood flow, blood volume, and off-resonance artifacts, and lack of direct validation in the tissue (40–42). In contrast, blood-based oximetry directly measures venous oxygen saturation, with the CS serving as a distinct vantage point similar to the invasive gold standard. Nevertheless, progress in CS oximetry has been hindered because of technical barriers (24). The need for complex in vitro calibration and prolonged breath-hold or navigator-gated scans made it unsuitable in the clinical setting and limited its clinical translation.

In particular, the CS poses specific challenges for quantitative oximetry because it is a mobile vessel that undergoes marked displacement (60% difference in displacement between the diastolic and

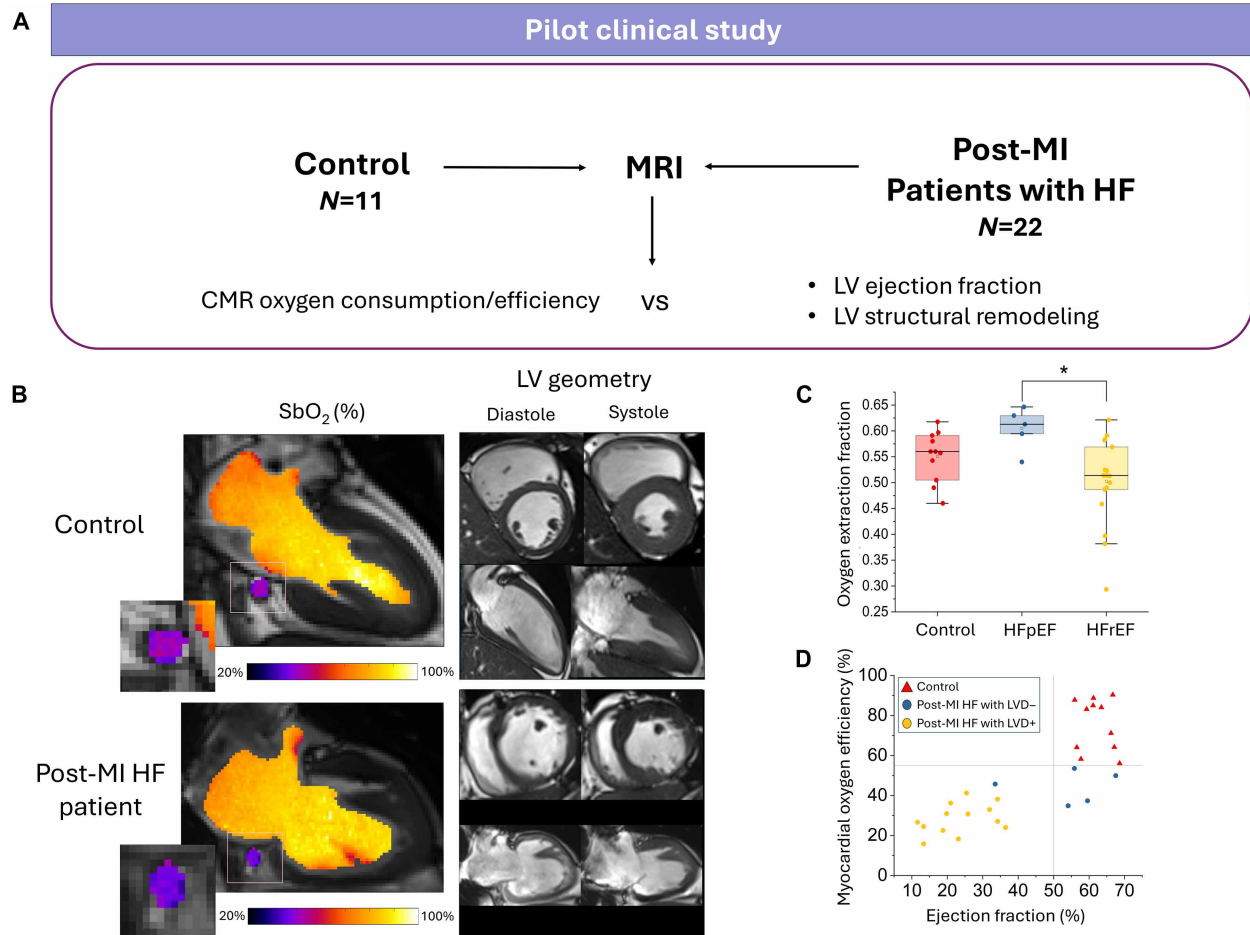


Fig. 4. CMR-based oxygen consumption and efficiency can reveal myocardial metabolic impairment in patients with HF. (A) The clinical study flow chart involves enrolling patients with HF who had MI and controls to analyze CMR oxygen consumption and efficiency versus LV heart function and structural remodeling. (B) Representative SbO₂ maps and cine imaging from one control participant and one patient with HF who had MI. The color scale indicates SbO₂: 20 to 100%. (C) The box plot compares OEF among the control group ($n = 11$), HFpEF group ($n = 5$), and HFrEF group ($n = 17$). The middle line in the boxes represents the median. The lower and upper bounds of the boxes represent the 25th and 75th percentiles, respectively. The whiskers represent the minimum and maximum values. Differences between conditions were analyzed using a one-way ANOVA followed by Fisher's least significant difference post hoc test. $*P < 0.05$. (D) The scatter plot shows MOE versus EF in control participants ($n = 11$), patients with HF and LVD- who had MI ($n = 5$), and patients with HF and LVD+ who had MI ($n = 13$).

systolic phases) throughout the cardiac cycle, with marked interindividual variability in the position and quiescent window (43). Whereas it is visible on cine images, these dynamics make the conventional breath-hold T_2/T_2^* mapping techniques, which are vulnerable to motion artifacts and partial-volume bias, unreliable in the CS (44). Our framework directly addresses these challenges with a continuous, free-breathing, motion-resolved 3D acquisition that retrospectively sorts data across cardiac and respiratory phases, ensuring stable CS tracking despite its complex motion and size variation (45, 46). Optimized BOLD contrast to noise combined with multiparametric T_1 - T_2 self-calibration improves measurement accuracy ($<0.1\%$ bias and $<2\%$ variability in simulations). Adiabatic pulses mitigate B_1 inhomogeneity, and gradient echo readouts reduce B_0 susceptibility, together improving robustness at 3 T. Moreover, the golden-angle radial trajectory with self-gating eliminates reliance on electrocardiography and breath-holds and is resistant to heart rate variability, as demonstrated in prior arrhythmia-resilient imaging studies (47–50)

These technical advancements ensure stable CS tracking and robust oximetry across the motion range, enabling reproducible global measurements of OEF in a rapid (<3 min), free-breathing scan. By eliminating the dependence on gating and calibration, our method is broadly applicable across diverse clinical populations, including those with arrhythmias, intolerance of breath-holding, or physiologic stress, for whom conventional gated acquisitions are unreliable. Beyond cardiac applications, this motion-robust, self-calibrated framework holds promise for extending oxygen consumption imaging to other moving organs such as the liver and kidney, addressing a broader unmet need in metabolic imaging. In the current study, to avoid potential bias in the LV blood (not in the targeted CS SbO₂ range), we used an infrared pulse blood oximetry probe to measure the SpO₂% for OEF derivation, which has been validated in prior studies and demonstrates high concordance with invasive central arterial measurements (51). However, in cases where prolonged breath holds are needed, e.g., gas challenge exams, the peripheral SpO₂% may experience time delays

Table 2. Demographic and physiological data of controls, post-MI HFpEF group, and post-MI HFrEF group. Among the 22 patients with HF who had MI, 4 patients were excluded from the analysis of MBF, MBF index, global MVO₂, and MOE because of technical limitations in CS blood flow measurement. The data are presented as the means (standard deviation). Dash entry indicates not applicable due to lack of data. Blank entries indicate that the sample size was too small to conduct statistical examination. SBP, systolic blood pressure; DBP, diastolic blood pressure; MBP, mean blood pressure; E/e', ratio of early mitral inflow velocity to early diastolic mitral annular velocity measured by echocardiography.

	Control	Post-MI HFpEF	Post-MI HFrEF	P
	n = 11	n = 5	n = 17	
Age	63.82 (11.57)	68.60 (5.22)	58.06 (10.46)	0.108
		Sex		
Male	10 (90.91%)	4 (80%)	15 (88.24%)	0.000
Female	1 (9.09%)	1 (20%)	2 (11.76%)	
		Laboratory data		
SBP (mmHg)	136.73 (7.00)	124.40 (9.53)	117.29 (21.44)*	0.018
DBP (mmHg)	81.45 (7.16)	77.00 (12.02)	74.06 (14.06)	0.290
MBP (mmHg)	98.40 (6.81)	92.80 (9.40)	89.63 (14.71)	0.183
HR (per min)	66.17 (8.02)	60.68 (13.66)	69.21 (12.11)	0.326
LV mass	91.03 (14.66)	122.95 (52.92)	138.64 (39.94)*	0.007
LV mass index	50.63 (6.45)	64.63 (23.37)	75.23 (20.55)*	0.005
Stroke volume	78.28 (12.42)	89.78 (35.44)	61.58 (17.16)†	0.012
Stroke volume index	43.64 (6.10)	47.37 (15.53)	33.77 (10.20)*,†	0.011
EF	62.18 (4.52)	60.58 (5.95)	26.27 (8.92)*,†	0.000
LVEDV (ml)	126.34 (21.27)	146.42 (48.34)	249.99 (72.56)*,†	0.000
LVEDV index (ml/m ²)	70.34 (9.81)	77.52 (20.71)	135.06 (33.92)*,†	0.000
LVESV (ml)	48.06 (11.48)	56.64 (16.00)	185.70 (69.29)*,†	0.000
LVESV index (ml/m ²)	26.69 (5.64)	30.15 (7.19)	99.99 (33.13)*,†	0.000
E/e'	–	11.12 (2.97)	14.79 (7.08)	
		Left ventricular metabolic profile		
MBF (ml/min)	61.30 (10.83)	67.24 (19.34)	80.25 (33.81)	0.196
MBF index (ml/g per minute)	0.69 (0.17)	0.68 (0.16)	0.62 (0.29)	0.759
Coronary sinus Sbo ₂ (%)	43.95 (4.26)	38.55 (3.81)	48.49 (8.19)†	0.016
OEF	0.55 (0.05)	0.60 (0.04)	0.50 (0.08)†	0.015
Global MVO ₂ (ml/min)	6.67 (1.31)	7.85 (2.20)	8.16 (3.39)	0.38
Stroke work (J)	99.08 (17.31)	67.96 (17.64)	45.73 (18.00)*,†	0.000
MOE (%)	75.68 (13.11)	43.92 (9.18)†	29.62 (8.63)*	0.000

*P < 0.05 for comparisons between the control group and the post-MI HFrEF group. †P < 0.05 for comparisons between the post-MI HFpEF group and the post-MI HFrEF group. ‡P < 0.05 for comparisons between the control group and the post-MI HFpEF group.

(52), where temporal corrections should be considered. Our current protocol also relies on a standard 2D phase contrast sequence to measure CS flow and derive MBF. Prior preclinical and clinical studies have established this method to measure the MBF of the whole heart without using contrast agents (53, 54). Although we found measures comparable to the invasive ground truth in our animal study, estimating MBF in the CS is relatively coarse compared with other voxel-wised MBF methods (53). Several factors may contribute to the moderate correlation observed between CMR and the invasive measurement of MBF, including the position sensitivity of CS thermodilution (55). Although we attempted to place both the CS catheter and CMR imaging plane at the same position of the CS (~3 cm from the ostium), deviations were unavoidable in some cases. Prior studies have shown that thermodilution-measured MBF can vary by up to 50% with only a 1-cm positional shift, whereas CS-SbO₂ remains relatively stable in this

position (24). Second, CMR-derived MBF was acquired under breath-holding conditions, whereas invasive measurements were obtained during free breathing. Breath-holding can alter intrathoracic pressure and coronary blood oxygen saturation, potentially influencing coronary flow (56, 57). These physiological differences may contribute to measurement variability between techniques. To address this issue, a free-breathing phase contrast is needed for inclusion in future studies. In addition, methods such as first-pass perfusion and arterial-spin labeling can be considered to further refine the MBF measurement and increase MVO₂ accuracy (58, 59). Despite these methodological differences, correlations remained robust and clinically meaningful, supporting the feasibility and reliability of our approach.

Besides MRI, cardiac metabolic imaging with ¹¹C-acetate PET has been developed to quantify MVO₂ through the tricarboxylic acid cycle and has shown promising results in human studies (2, 30, 60, 61).

However, its clinical scalability is limited by the tracer's short half-life and dependence on an on-site cyclotron. Similarly, ^{31}P MR spectroscopy and ^{13}C hyperpolarized MRI provide valuable metabolic insights but require specialized hardware and require major modification of the standard clinical workflows, restricting their use to research settings (62, 63). A further innovation of our study is the integration of high-resolution cine CMR to estimate cardiac work. In PET-based myocardial external efficiency studies, external work is typically computed from stroke volume and arterial pressure (mean arterial pressure or end-systolic pressure) and divided by PET-derived oxygen consumption. Although powerful, this method assumes simplified pressure profiles and lacks beat-wise temporal details. The clinical cine CMR provides gold-standard LV volumes [end-diastolic volume (EDV), end-systolic volume (ESV), and stroke volume] across the cardiac cycle. Combined with brachial cuff pressure and a modeled LV pressure waveform, these data enable noninvasive reconstruction of pressure-volume loops and derivation of stroke work (64). Recent validation studies have confirmed the accuracy of noninvasive pressure-volume loop estimation against invasive measurements in animals (65) and patients with HF (66). Thus, it offers a validated and clinically feasible foundation for estimating mechanical work, which we integrate with OEF and MBF to compute MOE noninvasively.

Our result is clinically relevant because myocardial ischemia remains the leading cause of HF in the developed world (67). Recent clinical trials have demonstrated that therapies enhancing glucose metabolism can improve MOE and cardiovascular outcomes, underscoring metabolic modulation as a promising target in HF management (4, 68). This highlights the need to quantify myocardial oxygen utilization as a biomarker for early intervention, disease monitoring, individualized therapy, and drug development. Our findings in patients with HF secondary to MI show that MOE may serve as an early indicator of cardiac dysfunction before EF reduction. A previous study reported that preserved EF is a common scenario in the subclinical HF population and showed an indifferent risk of death and hospitalization from patients with HFrEF (69). In our study, we observed that MOE had the potential to complement the left ventricular EF and detect energy efficiency during myocardial remodeling after ischemic myocardial injury (70). This inefficiency emerges before detectable structural or functional decline, making it a potential early marker for risk stratification in patients with HF.

Although our findings demonstrate promising results, several limitations of the study should be considered. First, our study performed the CMR and invasive measurements under different anesthesia sections because of procedural limitations. The cardiac output measured in both studies was stable and similar in RPP. However, a follow-up study to investigate the difference between varying anesthetics would be of interest. Second, peripheral blood pressure is measured via a cuff during MRI, which can introduce bias in cases of low blood pressure. Although our invasive study has shown no discrepancies from invasive measurements, more innovative blood pressure-measuring techniques may be adopted to further enhance accuracy in MOE quantification (71). Last, the clinical study was conducted with a relatively small sample size without performing stress tests and a lack of broader validation with PET and was limited to a single center. Expanding the study to include a larger patient cohort with PET and invasive validations from multiple centers and incorporating stress agents, such as adenosine or dobutamine, to evaluate the myocardial oxygen extraction reservoir could enhance the assessment of measurement reliability across different scanners and institutions.

The proposed method presents a practical approach for routinely assessing MVO₂ and efficiency in clinical settings. It can facilitate longitudinal studies and establish cardiac oxygen metabolism as a fundamental parameter in clinical decision-making (30). For example, maximal whole-body oxygen uptake (VO₂ max) is the standard reference for gauging the exercise capability of patients (72). VO₂ max is considered an index to measure integrated cardiac, pulmonary, and muscular functions, guiding the timing of heart transplantation and left ventricular assist device (7). The developed technique provides a complementary, heart-specific metric of oxygen consumption, enabling targeted cardiac metabolic risk stratification and treatment in patients with underlying heart diseases and facilitating earlier intervention to improve clinical outcomes. Furthermore, having an accessible and noncontrast tool to measure cardiac efficiency can reveal insights into cardiac metabolic research (73), with the ability to study preclinical individuals without exposing them to procedural risks. For example, myosin inhibitors (e.g., mavacamten), which the US Food and Drug Administration approves to treat obstructive hypertrophic cardiomyopathy (HCM), have transformed HCM care (74–76). Major interest has been shown in its application in nonobstructive individuals to slow down or prevent disease progression. However, the risk of overrelaxing the heart (77) in patients without myosin overaction has been a major concern. An accurate measure of oxygen consumption in patients with HCM may serve as a window to assess myosin function and guide treatment (78). In addition, such a tool can also aid in drug development by evaluating the effects of new cardiovascular drugs on cardiac metabolism (79). For example, recent clinical trials have explored pharmacologic interventions that enhance glucose metabolism and improve MOE, such as SGLT2 (sodium/glucose cotransporter 2) inhibitors and GLP-1 (glucagon-like peptide 1) receptor agonists. They demonstrate high promise in increasing total exercise capacity (68) and improving clinical outcomes (80–84). An accessible and fully noninvasive tool can shed light on the underlying mechanisms of these treatments and identify potential pathways for metabolically targeted therapies.

In addition to the blood oxygenation measurements, looking forward, our motion-resolved BOLD framework can also be applied to tissue-based quantification. By simultaneously capturing motion-resolved BOLD signals and anatomy, it could be combined with multiparametric tissue oximetry (e.g., T_2 , T_2^* , and susceptibility mapping) (21, 85–87) to improve spatial coverage and generate high-fidelity regional OEF and MVO₂ maps. Complementary noncontrast or first-pass perfusion techniques can further enhance supply-side measurements. At the same time, dynamic imaging throughout the cardiac cycle enables the assessment of regional mechanical efficiency (MOE). Together, this pathway opens the potential for reproducible regional quantification of OEF, MVO₂, and MOE, advancing the characterization of focal lesions. By enabling a fast, noncontrast, non-ionizing radioactive method for measuring cardiac oxygen metabolism, it can unlock frontiers for early diagnosis, personalized therapy, and the development of next-generation cardiometabolic treatments to combat the global HF epidemic.

MATERIALS AND METHODS

Study design

The goal of our study is to develop and validate rapid, motion-resolved MR oximetry for quantifying whole-heart myocardial oxygen extraction and efficiency and to evaluate its translational performance in

clinical settings. We followed a translational research paradigm spanning three sequential stages: numerical simulations for sequence optimization, validation against invasive reference standards in pigs, and prospective clinical evaluation in humans. The primary objective was to determine the accuracy and precision of MRI-derived CS blood oxygenation measurements compared with invasive CS catheterization. Secondary objectives included noninvasive quantification of global MVO_2 and MOE, with subsequent comparison between healthy controls and patients with HF secondary to MI. Sample numbers for each experimental component are specified below and are reported in the corresponding figure legends. Each individual contributed a single independent dataset to the analysis; no technical replicates were included. The image processing and data analysis were blindly performed without knowledge of the human participants' information. The human study was reported in accordance with the TREND reporting guidelines, and the checklist is provided in the Supplementary Materials.

Animal study

For the animal study, a total of 10 farm pigs (~28 to 35 kg) were approved by the Institutional Animal Care and Use Committee (IACUC no. 5257) and followed the ARRIVE (Animal Research: Reporting of In Vivo Experiments) guidelines. All animals were studied with CMR scans, followed by an invasive validation study within 1 week. Before the MRI scan and invasive CS catheterization, animals were prepared and underwent general anesthesia with mechanical ventilation at the baseline and adenosine administration (0.42 mg/kg per minute). The experiment was independently replicated in 10 pigs to evaluate reproducibility. Each animal underwent repeated imaging acquisitions to assess intraindividual measurement variability. The end point of this study was the agreement between MRI-derived CS blood oxygenation and invasive reference measurements.

Human study

For the human study, a prospective human study was conducted in a single institution under the institutional research board approval from February 2024 to February 2025 (IRB no. 2024-112-2), and every individual participated with informed consent. The electronic medical record was reviewed by a senior physician to screen eligible participants with an admission history of acute MI (88). Patients were enrolled on the basis of the following criteria: inclusion: MI diagnosis at least 1 month before MRI and current or prior symptoms or signs of HF; exclusion: active devices (pacemakers and implantable cardioverter-defibrillators), structural heart diseases, and claustrophobia. Age- and sex-matched individuals without any known cardiovascular diseases, with no smoking history, and with no family history of premature coronary heart disease were included in the control group.

The patient cohort was classified into the HFrEF group and HFpEF group on the basis of $\text{EF} \leq 40\%$ and $\text{EF} \geq 50\%$, respectively. Individuals with EF between 40 and 50% were not enrolled in the study. Furthermore, the patient cohort was divided into two categories on the basis of the presentation of dilated LV. Patients with (post-MI HF with LVD+) and without (post-MI HF with LVD-) adverse left ventricular remodeling were categorized using the upper normal limit according to age- and sex-normalized reference standards published in the literature (36). The end points included the between-group comparison of MRI-derived myocardial oxygen extraction, MBF, MVO_2 , and MOE. In a separate set of patients ($N = 3$), patients with patent foramen ovale, pulmonary hypertension, and atrial septal defect were

recruited and went through right heart catheterization for validation of CS blood oxygenation.

CS catheterization

Invasive CS catheterization was performed in five pigs using a Swan-Ganz catheter (Edwards Lifesciences Japan Ltd.) at the baseline and following the administration of adenosine (0.42 mg/kg per minute) within 1 week of the CMR scans. Vascular access was established in the internal carotid artery and internal jugular vein. Under fluoroscopic guidance, the Swan-Ganz catheter was advanced into the right atrium and CS via the internal jugular vein. Blood samples were drawn from both the right atrium and CS for immediate analysis of oxygen saturation and Hct using i-STAT CG8+ cartridges (Abbott Laboratories). CS blood flow was measured by thermodilution through the injection of 5 ml of ice-cold saline into the catheter positioned in the CS (89). The average of three measurements was used for the final analysis. Arterial blood samples were obtained from the carotid artery to determine arterial oxygen saturation. The invasive measurements of CS blood flow, Hct, and blood oxygen saturation served as the reference (ground truth) for the proposed MR technique.

CMR imaging protocol

All animals were scanned using a Clinical 3.0-T scanner (Vida, Siemens Healthineer AG) at Cedars Sinai Medical Center. Human participants were scanned using a clinical 3.0-T scanner (MAGNETOM Vida, Siemens Healthineer AG) at the First Hospital of China Medical University. The same imaging parameters are adopted at both sites. Cardiac MR scans with the same imaging parameters were performed in both animals and human studies (the details of imaging acquisition parameters and data analysis are listed in the Supplementary Materials). After localization, a clinical 2D cine sequence was prescribed to visualize the heart anatomy and measure the heart function. The CS was identified in the arterioventricular groove using the basal slices of the short-axis cine images. The cross-sectional view for flow and oximetry measurement was positioned perpendicular to CS at 1 to 2 cm from the ostium (90). In the human study, late gadolinium enhancement images were collected to confirm the MIs in the patients.

During the imaging acquisition, the heart rate (HR), blood pressure, and arterial blood oxygen saturation (SpO_2) were recorded using clinical equipment. The image processing and data analysis were blindly performed without knowledge of the participants' information. Outliers were identified using predefined statistical criteria (>3 standard deviations from the mean). No data points were excluded based solely on extreme outcome values.

Statistical analysis

The data are presented as the means $\pm$ standard deviations for continuous variables and as counts and percentages for categorical variables. Bland-Altman analysis was performed to assess the agreement between cardiac MRI and invasive measurements of MBF, OEF, and global MVO_2 . Simple linear regression was used to estimate the correlation between the two methods. To compare MBF, OEF, global MVO_2 , and MOE between the baseline and adenosine status, paired Student's t test was used. For group comparisons among control, post-MI with LVD+, and post-MI with LVD- in the human study, a one-way analysis of variance (ANOVA) with Tukey post hoc analysis was applied. A two-sided P value of <0.05 was considered statistically significant. All statistical analyses were performed using IBM

SPSS (version 24.0, IBM Corporation). All individual level data for $n < 20$ are presented in data file S1. All reconstruction model fitting and nonclinical postprocessing were performed with self-developed MATLAB scripts on a workstation equipped with two 2.70-GHz 12-core Intel Xeon central processing units, an NVIDIA 4090 graphics processing unit, and 128 gigabytes of random-access memory.

Supplementary Materials

The PDF file includes:

Materials and Methods

Figs. S1 to S5

Table S1

Trend Statement Checklist

References (91–97)

Other Supplementary Material for this manuscript includes the following:

Data file S1

MDAR Reproducibility Checklist

REFERENCES AND NOTES

- G. D. Lopuschuk, Q. G. Karwi, R. Tian, A. R. Wende, E. D. Abel, Cardiac energy metabolism in heart failure. *Circ. Res.* **128**, 1487–1513 (2021).
- R. Y. Parbhudayal, H. J. Harms, M. Michels, A. C. van Rossum, T. Germans, J. van der Velden, Increased myocardial oxygen consumption precedes contractile dysfunction in hypertrophic cardiomyopathy caused by pathogenic *TNNI2* gene variants. *J. Am. Heart Assoc.* **9**, e015316 (2020).
- J. Roibal Pravio, E. Barge Caballero, C. Barbeito Caamaño, M. J. Paniagua Martin, G. Barge Caballero, D. Couto Mallon, P. Pardo Martinez, Z. Grille Cancela, P. Blanco Canosa, J. M. García Pinilla, J. M. Vázquez Rodríguez, M. G. Crespo Leiro, Determinants of maximal oxygen uptake in patients with heart failure. *ESC Heart Fail.* **8**, 2002–2008 (2021).
- G. D. Lopuschuk, S. Verma, Mechanisms of cardiovascular benefits of sodium glucose co-transporter 2 (SGLT2) inhibitors: A state-of-the-art review. *JACC Basic Transl. Sci.* **5**, 632–644 (2020).
- M. J. Previs, T. S. O'Leary, M. P. Morley, B. M. Palmer, M. LeWinter, J. M. Yob, F. D. Pagani, C. Petucci, M. S. Kim, K. B. Margulies, Z. Arany, D. P. Kelly, S. M. Day, Defects in the proteome and metabolome in human hypertrophic cardiomyopathy. *Circ. Heart Fail.* **15**, e009521 (2022).
- S. Monga, L. Valković, S. G. Myerson, S. Neubauer, M. Mahmood, O. J. Rider, Role of cardiac energetics in aortic stenosis disease progression: Identifying the high-risk metabolic phenotype. *Circ. Cardiovasc. Imaging* **16**, e014863 (2023).
- P. A. Heidenreich, B. Bozkurt, D. Aguilar, L. A. Allen, J. J. Byun, M. M. Colvin, A. Deswal, M. H. Drazner, S. M. Dunlay, L. R. Evers, J. C. Fang, S. E. Fedson, G. C. Fonarow, S. S. Hayek, A. F. Hernandez, P. Khazanie, M. M. Kittleson, C. S. Lee, M. S. Link, C. A. Milano, L. C. Natcheta, A. T. Sandhu, L. W. Stevenson, O. Vardeny, A. R. Vest, C. W. Yancy, ACC/AHA Joint Committee Members, 2022 AHA/ACC/HFSA guideline for the management of heart failure: A report of the American College of Cardiology/American Heart Association joint committee on clinical practice guidelines. *Circulation* **145**, e895–e1032 (2022).
- N. Marzlin, A. G. Hays, M. Peters, A. Kaminski, S. Roemer, P. O'Leary, S. Kroboth, D. R. Harland, B. K. Khandheria, A. J. Tajik, R. Jain, Myocardial work in echocardiography. *Circ. Cardiovasc. Imaging* **16**, e014419 (2023).
- L. C. Boyette, B. Manna, "Physiology, myocardial oxygen demand," in *StatPearls* (StatPearls, Treasure Island, 2023).
- B. M. Everett, M. V. Moorthy, J. T. Tikkanen, N. R. Cook, C. M. Albert, Markers of myocardial stress, myocardial injury, and subclinical inflammation and the risk of sudden death. *Circulation* **142**, 1148–1158 (2020).
- P. Knaapen, T. Germans, J. Knuuti, W. J. Paulus, P. A. Dijkmans, C. P. Allaart, A. A. Lammersma, F. C. Visser, Myocardial energetics and efficiency: Current status of the noninvasive approach. *Circulation* **115**, 918–927 (2007).
- D. Agostini, H. Iida, A. Takahashi, Y. Tamura, M. Henry Amar, Y. Ono, Regional myocardial metabolic rate of oxygen measured by O2-15 Inhalation and positron emission tomography in patients with cardiomyopathy. *Clin. Nucl. Med.* **26**, 41–49 (2001).
- S. R. Yurista, R. A. Eder, D. H. Kwon, C. T. Farrar, Y. F. Yen, W. H. W. Tang, C. T. Nguyen, Magnetic resonance imaging of cardiac metabolism in heart failure: How far have we come? *Eur. Heart J. Cardiovasc. Imaging* **23**, 1277–1289 (2022).
- G. Brix, E. A. Nekolla, D. Nosske, J. Griebel, Risks and safety aspects related to PET/MR examinations. *Eur. J. Nucl. Med. Mol. Imaging* **36**, S131–S138 (2009).
- D. J. Heeger, D. Ress, What does fMRI tell us about neuronal activity? *Nat. Rev. Neurosci.* **3**, 142–151 (2002).
- J. W. Belliveau, D. N. Kennedy Jr., R. C. McKinstry, B. R. Buchbinder, R. M. Weisskoff, M. S. Cohen, J. M. Vevea, T. J. Brady, B. R. Rosen, Functional mapping of the human visual cortex by magnetic resonance imaging. *Science* **254**, 716–719 (1991).
- S. Ogawa, T. M. Lee, A. R. Kay, D. W. Tank, Brain magnetic resonance imaging with contrast dependent on blood oxygenation. *Proc. Natl. Acad. Sci. U.S.A.* **87**, 9868–9872 (1990).
- K. R. Thulborn, J. C. Waterton, P. M. Matthews, G. K. Radda, Oxygenation dependence of the transverse relaxation time of water protons in whole blood at high field. *Biochim. Biophys. Acta* **714**, 265–270 (1982).
- K. Fischer, D. P. Guensch, B. Jung, I. King, H. von Tengg-Kobligk, N. Giannetti, B. Eberle, M. G. Friedrich, Insights into myocardial oxygenation and cardiovascular magnetic resonance tissue biomarkers in heart failure with preserved ejection fraction. *Circ. Heart Fail.* **15**, e008903 (2022).
- H. J. Yang, I. Oksuz, D. Dey, J. Sykes, M. Klein, J. Butler, M. S. Kovacs, O. Sobczyk, I. Cokic, P. J. Slomka, X. Bi, D. Li, M. Tighiouart, S. A. Tsiftaris, F. S. Prato, J. A. Fisher, R. Dharmakumar, Accurate needle-free assessment of myocardial oxygenation for ischemic heart disease in canines using magnetic resonance imaging. *Sci. Transl. Med.* **11**, eaat4407 (2019).
- L. Lu, C. Eldeniz, H. An, R. Li, Y. Yang, T. H. Schindler, L. R. Peterson, P. K. Woodard, J. Zheng, Quantification of myocardial oxygen extraction fraction: A proof-of-concept study. *Magn. Reson. Med.* **85**, 3318–3325 (2021).
- J. A. Stainsby, G. A. Wright, Monitoring blood oxygen state in muscle microcirculation with transverse relaxation. *Magn. Reson. Med.* **45**, 662–672 (2001).
- J. Varghese, L. C. Potter, R. LaFountain, X. Pan, S. V. Raman, R. Ahmad, O. P. Simonetti, CMR-based blood oximetry via multi-parametric estimation using multiple T2 measurements. *J. Cardiovasc. Magn. Reson.* **19**, 88 (2017).
- W. D. Foltz, N. Merchant, E. Downar, J. A. Stainsby, G. A. Wright, Coronary venous oximetry using MRI. *Magn. Reson. Med.* **42**, 837–848 (1999).
- G. A. Wright, B. S. Hu, A. Macovski, Estimating oxygen saturation of blood in vivo with MR imaging at 1.5 T. *J. Magn. Reson. Imaging* **1**, 275–283 (1991).
- Y. Wen, T. D. Nguyen, Z. Liu, P. Spincemille, D. Zhou, A. Dimov, Y. Kee, K. Deh, J. Kim, J. W. Weinsaft, Y. Wang, Cardiac quantitative susceptibility mapping (QSM) for heart chamber oxygenation. *Magn. Reson. Med.* **79**, 1545–1552 (2018).
- S. Portnoy, M. Seed, J. G. Sled, C. K. Macgowan, Non-invasive evaluation of blood oxygen saturation and hematocrit from T1 and T2 relaxation times: In-vitro validation in fetal blood. *Magn. Reson. Med.* **78**, 2352–2359 (2017).
- L. A. E. Brown, G. S. Gulsin, S. C. Oncul, D. A. Broadbent, J. L. Yeo, A. L. Wood, C. E. D. Saunderson, A. Das, N. Jex, A. Chowdhary, S. Thirunavukarasu, N. Sharrack, K. D. Knott, E. Levelt, P. P. Swoboda, H. Xue, J. P. Greenwood, J. C. Moon, D. Adlam, G. P. McCann, P. Kellman, S. Plein, Sex- and age-specific normal values for automated quantitative pixel-wise myocardial perfusion cardiovascular magnetic resonance. *Eur. Heart J. Cardiovasc. Imaging* **24**, 426–434 (2022).
- C. M. Kramer, J. Barkhausen, C. Bucciarelli-Ducci, S. D. Flamm, R. J. Kim, E. Nagel, Standardized cardiovascular magnetic resonance imaging (CMR) protocols: 2020 update. *J. Cardiovasc. Magn. Reson.* **22**, 17 (2020).
- J. Sörensen, H. J. Harms, J. M. Aalen, T. Baron, O. A. Smiseth, F. A. Flachskampf, Myocardial efficiency: A fundamental physiological concept on the verge of clinical impact. *JACC Cardiovasc. Imaging* **13**, 1564–1576 (2020).
- K. Stoschitzky, Blood pressure, heart rate, or the rate pressure product: What is the best predictor of clinical outcome? *Eur. Heart J. Open.* **2**, oeac063 (2022).
- W. K. Shen, Y. Kurachi, Mechanisms of adenosine-mediated actions on cellular and clinical cardiac electrophysiology. *Mayo Clin. Proc.* **70**, 274–291 (1995).
- S. R. Bulley, C. Wittnich, Adenosine infusion: A rational approach towards induced hypotension. *Can. J. Cardiol.* **11**, 327–334 (1995).
- A. Faragli, A. Alogna, C. B. Lee, M. Zhu, N. Ghorbani, F. P. Lo Muzio, B. Schnackenburg, C. Stehning, T. Kuehne, H. Post, L. Goubergrits, E. Nagel, B. Pieske, S. Kelle, M. Kelm, Non-invasive CMR-based quantification of myocardial power and efficiency under stress and ischemic conditions in Landrace pigs. *Front. Cardiovasc. Med.* **8**, 689255 (2021).
- B. Bozkurt, A. J. S. Coats, H. Tsutsui, C. M. Abdelhamid, S. Adamopoulos, N. Albert, S. D. Anker, J. Atherton, M. Böhm, J. Butler, M. H. Drazner, G. Michael Felker, G. Filippatos, M. Fluzat, G. C. Fonarow, J. E. Gomez-Mesa, P. Heidenreich, T. Imamura, E. A. Jankowska, J. Januzzi, P. Khazanie, K. Kinugawa, C. S. P. Lam, Y. Matsue, M. Metra, T. Ohtani, M. Francesco Piepoli, P. Ponikowski, G. M. C. Rosano, Y. Sakata, P. Seferović, R. C. Starling, J. R. Teerlink, O. Vardeny, K. Yamamoto, C. Yancy, J. Zhang, S. Zieroth, Universal definition and classification of heart failure: A report of the Heart Failure Society of America, Heart Failure Association of the European Society of Cardiology, Japanese Heart Failure Society and Writing Committee of the Universal Definition of Heart Failure. *Eur. J. Heart Fail.* **23**, 352–380 (2021).
- A. M. Maceira, S. K. Prasad, M. Khan, D. J. Pennell, Normalized left ventricular systolic and diastolic function by steady state free precession cardiovascular magnetic resonance. *J. Cardiovasc. Magn. Reson.* **8**, 417–426 (2006).

37. D. S. Fieno, S. M. Shea, Y. Li, K. R. Harris, J. P. Finn, D. Li, Myocardial perfusion imaging based on the blood oxygen level-dependent effect using T2-prepared steady-state free-precession magnetic resonance imaging. *Circulation* **110**, 1284–1290 (2004).
38. S. A. Tsaftaris, X. Zhou, R. Tang, D. Li, R. Dharmakumar, Detecting myocardial ischemia at rest with cardiac phase-resolved blood oxygen level-dependent cardiovascular magnetic resonance. *Circ. Cardiovasc. Imaging* **6**, 311–319 (2013).
39. C. Jahnke, R. Gebker, R. Manka, B. Schnackenburg, E. Fleck, I. Paetsch, Navigator-gated 3D blood oxygen level-dependent CMR at 3.0-T for detection of stress-induced myocardial ischemic reactions. *JACC Cardiovasc. Imaging* **3**, 375–384 (2010).
40. J. J. Weyers, V. Ramanan, A. Javed, J. Barry, M. Larsen, K. Nayak, G. A. Wright, N. R. Ghugre, Myocardial blood flow is the dominant factor influencing cardiac magnetic resonance adenosine stress T2. *NMR Biomed.* **35**, e4643 (2022).
41. A. T. O'Brien, K. E. Gil, J. Varghese, O. P. Simonetti, K. M. Zareba, T2 mapping in myocardial disease: A comprehensive review. *J. Cardiovasc. Magn. Reson.* **24**, 33 (2022).
42. M. G. Friedrich, T. D. Karamitsos, Oxygenation-sensitive cardiovascular magnetic resonance. *J. Cardiovasc. Magn. Reson.* **15**, 43 (2013).
43. J. D. Suever, P. J. Watson, R. L. Eisner, S. Lerakis, R. E. O'Donnell, J. N. Oshinski, Time-resolved analysis of coronary vein motion and cross-sectional area. *J. Magn. Reson. Imaging* **34**, 811–815 (2011).
44. J. Varghese, M. Smyke, Y. Pan, S. Rajpal, J. Craft, L. C. Potter, S. V. Raman, R. Ahmad, O. P. Simonetti, Patient-adaptive magnetic resonance oximetry: Comparison with invasive catheter measurement of blood oxygen saturation in patients with cardiovascular disease. *J. Magn. Reson. Imaging* **52**, 1449–1459 (2020).
45. S. Miller, O. P. Simonetti, J. Carr, U. Kramer, J. P. Finn, MR imaging of the heart with cine true fast imaging with steady-state precession: Influence of spatial and temporal resolutions on left ventricular functional parameters. *Radiology* **223**, 263–269 (2002).
46. J. Dou, T. G. Reese, W. Y. Tseng, V. J. Wedeen, Cardiac diffusion MRI without motion effects. *Magn. Reson. Med.* **48**, 105–114 (2002).
47. A. G. Christodoulou, J. L. Shaw, C. Nguyen, Q. Yang, Y. Xie, N. Wang, D. Li, Magnetic resonance multitasking for motion-resolved quantitative cardiovascular imaging. *Nat. Biomed. Eng.* **2**, 215–226 (2018).
48. F. Contijoch, S. K. Iyer, J. J. Pilla, P. Yushkevich, J. H. Gorman III, R. C. Gorman, H. Litt, Y. Han, W. R. T. Witschey, Self-gated MRI of multiple beat morphologies in the presence of arrhythmias. *Magn. Reson. Med.* **78**, 678–688 (2017).
49. Z. Ming, A. Pogoyan, A. G. Christodoulou, J. P. Finn, D. Ruan, K. L. Nguyen, Dynamic regularized adaptive cluster optimization (DRACO) for quantitative cardiac cine MRI in complex arrhythmias. *J. Magn. Reson. Imaging* **61**, 248–262 (2025).
50. L. Feng, L. Axel, H. Chandarana, K. T. Block, D. K. Sodickson, R. Otazo, XD-GRASP: Golden-angle radial MRI with reconstruction of extra motion-state dimensions using compressed sensing. *Magn. Reson. Med.* **75**, 775–788 (2016).
51. G. Gok, T. Cinar, Z. Nurkalem, D. Duman, Correlation of pulse oximetry oxygen saturation with blood gas arterial oxygen saturation in patients with heart failure reduced ejection fraction: Prospective cohort study. *Ann. Med. Res.* **27**, 1073–1076 (2021).
52. J. W. Severinghaus, J. F. Kelleher, Recent developments in pulse oximetry. *Anesthesiology* **76**, 1018–1038 (1992).
53. J. K. Tingsgaard, M. H. Sørensen, A. S. Bojer, R. H. Anderson, D. A. Broadbent, S. Plein, P. Gæde, P. L. Madsen, Myocardial blood flow determination from contrast-free magnetic resonance imaging quantification of coronary sinus flow. *J. Magn. Reson. Imaging* **59**, 1258–1266 (2024).
54. J. W. Koskenvuo, H. Sakuma, P. Niemi, J. O. Toikka, J. Knuuti, H. Laine, M. Komu, M. Korman, M. Saraste, J. J. Hartiala, Global myocardial blood flow and global flow reserve measurements by MRI and PET are comparable. *J. Magn. Reson. Imaging* **13**, 361–366 (2001).
55. J. P. Bagger, Coronary sinus blood flow determination by the thermol dilution technique: Influence of catheter position and respiration. *Cardiovasc. Res.* **19**, 27–31 (1985).
56. M. Ferrigno, D. D. Hickey, M. H. Linér, C. E. Lundgren, Cardiac performance in humans during breath holding. *J. Appl. Physiol.* **60**, 1871–1877 (1986).
57. M. Pelletier-Galarneau, R. A. deKemp, C. R. R. N. Hunter, R. Klein, M. Klein, J. Ironstone, J. A. Fisher, T. D. Ruddy, Effects of hypercapnia on myocardial blood flow in healthy human subjects. *J. Nucl. Med.* **59**, 100–106 (2018).
58. R. Catania, S. Quinn, A. A. Rahsepar, T. Agirlar Trabzonlu, J. B. Bisen, K. Chow, D. C. Lee, R. Avery, P. Kellman, B. D. Allen, Quantitative stress first-pass perfusion cardiac MRI: State of the art. *Radiographics* **45**, e240115 (2025).
59. F. Kober, T. Jao, T. Troalen, K. S. Nayak, Myocardial arterial spin labeling. *J. Cardiovasc. Magn. Reson.* **18**, 22 (2016).
60. S. A. Timmer, T. Germans, M. J. Götte, I. K. Rüssel, P. A. Dijkman, M. Lubberink, J. M. ten Berg, F. J. ten Cate, A. A. Lammertsma, P. Knaapen, A. C. van Rossum, Determinants of myocardial energetics and efficiency in symptomatic hypertrophic cardiomyopathy. *Eur. J. Nucl. Med. Mol. Imaging* **37**, 779–788 (2010).
61. T. S. Clemmensen, J. Soerensen, N. H. Hansson, L. P. Tolbod, H. J. Harms, H. Eiskjær, F. Mikkelsen, H. Wiggers, N. F. Andersen, S. H. Poulsen, Myocardial oxygen consumption and efficiency in patients with cardiac amyloidosis. *J. Am. Heart Assoc.* **7**, e009974 (2018).
62. S. Neubauer, T. Krahe, R. Schindler, M. Horn, H. Hillenbrand, C. Entzeroth, H. Mader, E. P. Kromer, G. A. Riegger, K. Lackner, ³¹P magnetic resonance spectroscopy in dilated cardiomyopathy and coronary artery disease. Altered cardiac high-energy phosphate metabolism in heart failure. *Circulation* **86**, 1810–1818 (1992).
63. O. J. Rider, D. J. Tyler, Clinical implications of cardiac hyperpolarized magnetic resonance imaging. *J. Cardiovasc. Magn. Reson.* **15**, 93 (2013).
64. H. Suga, Total mechanical energy of a ventricle model and cardiac oxygen consumption. *Am. J. Physiol.* **236**, H498–H505 (1979).
65. F. Seemann, P. Arvidsson, D. Nordlund, S. Kopic, M. Carlsson, H. Arheden, E. Heiberg, Noninvasive quantification of pressure-volume loops from brachial pressure and cardiovascular magnetic resonance. *Circ. Cardiovasc. Imaging* **12**, e008493 (2019).
66. P. M. Arvidsson, P. G. Green, W. D. Watson, M. Shanmuganathan, E. Heiberg, G. L. De Maria, H. Arheden, N. Herring, O. J. Rider, Non-invasive left ventricular pressure-volume loops from cardiovascular magnetic resonance imaging and brachial blood pressure: Validation using pressure catheter measurements. *Eur. Heart J. Imaging Methods Pract.* **1**, qyad035 (2023).
67. M. White, J. L. Rouleau, T. D. Ruddy, T. De Marco, D. Moher, K. Chatterjee, Decreased coronary sinus oxygen content: A predictor of adverse prognosis in patients with severe congestive heart failure. *J. Am. Coll. Cardiol.* **18**, 1631–1637 (1991).
68. L. Lee, J. Horowitz, M. Frenneaux, Metabolic manipulation in ischaemic heart disease, a novel approach to treatment. *Eur. Heart J.* **25**, 634–641 (2004).
69. S. Bhatia, J. V. Tu, D. S. Lee, P. C. Austin, J. Fang, A. Haouzi, Y. Gong, P. P. Liu, Outcome of heart failure with preserved ejection fraction in a population-based study. *N. Engl. J. Med.* **355**, 260–269 (2006).
70. T. Kato, S. Niizuma, Y. Inuzuka, T. Kawashima, J. Okuda, Y. Tamaki, Y. Iwanaga, M. Narazaki, T. Matsuda, T. Soga, T. Kita, T. Kimura, T. Shioi, Analysis of metabolic remodeling in compensated left ventricular hypertrophy and heart failure. *Circ. Heart Fail.* **3**, 420–430 (2010).
71. L. Wang, S. Tian, R. Zhu, A new method of continuous blood pressure monitoring using multichannel sensing signals on the wrist. *Microsyst. Nanoeng.* **9**, 117 (2023).
72. G. F. Fletcher, P. A. Ades, P. Kligfield, R. Arena, G. J. Balady, V. A. Bittner, L. A. Coke, J. L. Fleg, D. E. Forman, T. C. Gerber, M. Gulati, K. Madan, J. Rhodes, P. D. Thompson, M. A. Williams, American Heart Association Exercise, Cardiac Rehabilitation, and Prevention Committee of the Council on Clinical Cardiology, Council on Nutrition, Physical Activity and Metabolism, Council on Cardiovascular and Stroke Nursing, and Council on Epidemiology and Prevention, Exercise standards for testing and training. *Circulation* **128**, 873–934 (2013).
73. H. Taegtmeier, M. E. Young, G. D. Lopaschuk, E. D. Abel, H. Brunengraber, V. Darley-Usmar, C. Des Rosiers, P. A. Ades, J. F. Glatz, J. L. Griffin, R. J. Gropler, H. G. Holzhueter, J. R. Kizer, E. D. Lewandowski, C. R. Malloy, S. Neubauer, L. R. Peterson, M. A. Portman, F. A. Recchia, J. E. Van Eyk, T. J. Wang, American Heart Association Council on Basic Cardiovascular Sciences, Assessing cardiac metabolism: A scientific statement from the American Heart Association. *Circ. Res.* **118**, 1659–1701 (2016).
74. M. S. Maron, A. Masri, M. E. Nassif, R. Barriales-Villa, M. Arad, N. Cardim, L. Choudhury, B. Claggett, C. J. Coats, H. D. Düngen, P. Garcia-Pavia, A. A. Hagège, J. L. Januzzi, M. M. Y. Lee, G. D. Lewis, C. S. Ma, M. Michels, I. Olivetto, A. Oreziak, A. T. Owens, M. A. Spertus, S. D. Solomon, J. Tfelt-Hansen, M. van Sinttruij, J. Veselka, H. Watkins, D. L. Jacoby, S. B. Heitner, S. Kupfer, F. I. Malik, L. Meng, A. Wohltman, T. P. Abraham, SEQUOIA-HCM Investigators, Aficamten for symptomatic obstructive hypertrophic cardiomyopathy. *N. Engl. J. Med.* **390**, 1849–1861 (2024).
75. I. Olivetto, A. Oreziak, R. Barriales-Villa, T. P. Abraham, A. Masri, P. Garcia-Pavia, S. Saberi, N. K. Lakdawala, M. T. Wheeler, A. Owens, M. Kubanek, W. Wojakowski, M. K. Jensen, J. Gimeno-Blanes, K. Afshar, J. Myers, S. M. Hegde, S. D. Solomon, A. J. Sehnert, D. Zhang, W. Li, M. Bhattacharya, J. M. Edelberg, C. B. Waldman, S. J. Lester, A. Wang, C. Y. Ho, D. Jacoby, EXPLORER-HCM study investigators, Mavacamten for treatment of symptomatic obstructive hypertrophic cardiomyopathy (EXPLORER-HCM): A randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet* **396**, 759–769 (2020).
76. M. Y. Desai, A. Owens, K. Wolski, J. B. Geske, S. Saberi, A. Wang, M. Sherrid, P. C. Cremer, N. K. Lakdawala, A. Tower-Rader, D. Ferrin, S. S. Naidu, N. G. Smedira, H. Schaff, E. McElean, C. Sewell, L. Mudarris, Z. Gong, K. Lampl, A. J. Sehnert, S. E. Nissen, Mavacamten in patients with hypertrophic cardiomyopathy referred for septal reduction: Week 56 results from the VALOR-HCM randomized clinical trial. *JAMA Cardiol.* **8**, 968–977 (2023).
77. M. Y. Desai, D. Seto, M. Cheung, S. Afsari, N. Patel, A. Bastien, J. Lockman, M. Coiro, M. W. Martinez, Mavacamten: Real-world experience from 22 months of the risk evaluation and mitigation strategy (REMS) program. *Circ. Heart Fail.* **18**, e012441 (2025).
78. M. Y. Desai, S. E. Nissen, T. Abraham, I. Olivetto, P. Garcia-Pavia, R. D. Lopes, N. Verheyen, O. Wever-Pinzon, K. Wolski, W. Jaber, L. Mitchell, D. Davey, J. Myers, T. Rano, V. Bhatia, Y. Zhong, S. Carter-Bonanza, V. Florea, R. Aronson, A. T. Owens, ODYSSEY-HCM Investigators, Mavacamten in symptomatic nonobstructive hypertrophic cardiomyopathy: Design, rationale, and baseline characteristics of ODYSSEY-HCM. *JACC Heart Fail.* **13**, 358–370 (2025).

79. H. Honka, C. Solis-Herrera, C. Triplitt, L. Norton, J. Butler, R. A. DeFronzo, Therapeutic manipulation of myocardial metabolism: JACC state-of-the-art review. *J. Am. Coll. Cardiol.* **77**, 2022–2039 (2021).
80. J. Butler, W. S. Jones, J. A. Udell, S. D. Anker, M. C. Petrie, J. Harrington, M. Mattheus, I. Zwerner, O. Amir, M. C. Bahit, J. Bauersachs, A. Bayes-Genis, Y. Chen, V. K. Chopra, G. Figtree, J. Ge, S. G. Goodman, N. Gotcheva, S. Goto, T. Gasior, W. Jamal, J. L. Januzzi, M. H. Jeong, Y. Lopatin, R. D. Lopes, B. Merkely, P. B. Parikh, A. Parkhomenko, P. Ponikowski, X. Rossello, M. Schou, D. Simic, P. G. Steg, J. Szachniewicz, P. van der Meer, D. Vinereanu, S. Zieroth, M. Brueckmann, M. Sumin, D. L. Bhatt, A. F. Hernandez, Empagliflozin after acute myocardial infarction. *N. Engl. J. Med.* **390**, 1455–1466 (2024).
81. S. D. Solomon, J. J. V. McMurray, B. Claggett, R. A. de Boer, D. DeMets, A. F. Hernandez, S. E. Inzucchi, M. N. Kosiborod, C. S. P. Lam, F. Martinez, S. J. Shah, A. S. Desai, P. S. Jhund, J. Belohlavek, C. E. Chiang, C. J. W. Borliff, J. Comin-Colet, D. Dobreaanu, J. Drozd, J. C. Fang, M. A. Alcocer-Gamba, W. Al Habeey, Y. Han, J. W. Cabrera Honorio, S. P. Janssens, T. Katova, M. Kitakaze, B. Merkely, E. O'Meara, J. F. K. Saraiva, S. N. Tereshchenko, J. Thierer, M. Vaduganathan, O. Vardeny, S. Verma, V. N. Pham, U. Wilderang, N. Zozerska, E. Bachus, D. Lindholm, M. Petersson, A. M. Langkilde, DELIVER Trial Committees and Investigators, Dapagliflozin in heart failure with mildly reduced or preserved ejection fraction. *N. Engl. J. Med.* **387**, 1089–1098 (2022).
82. S. James, D. Erlinge, R. F. Storey, D. K. McGuire, M. de Belder, N. Eriksson, K. Andersen, D. Austin, G. Arefalk, D. Carrick, R. Hofmann, S. P. Hoole, D. A. Jones, K. Lee, H. Tygesen, P. A. Johansson, A. M. Langkilde, W. Ridderstråle, E. Parvaresh Rizi, J. Deanfield, J. Oldgren, Dapagliflozin in myocardial infarction without diabetes or heart failure. *NEJM Evid.* **3**, EVIDo2300286 (2024).
83. M. Packer, M. R. Zile, C. M. Kramer, S. J. Baum, S. E. Litwin, V. Menon, J. Ge, G. J. Weerakkody, Y. Ou, M. C. Bunck, K. C. Hurt, M. Murakami, B. A. Borlaug, SUMMIT Trial Study Group, Tirzepatide for heart failure with preserved ejection fraction and obesity. *N. Engl. J. Med.* **392**, 427–437 (2025).
84. A. M. Lincoff, K. Brown-Frandsen, H. M. Colhoun, J. Deanfield, S. S. Emerson, S. Esbjerg, S. Hardt-Lindberg, G. K. Hovingh, S. E. Kahn, R. F. Kushner, I. Lingvay, T. K. Oral, M. M. Michelsen, J. Plutzky, C. W. Tornøe, D. H. Ryan, SELECT Trial Investigators, Semaglutide and cardiovascular outcomes in obesity without diabetes. *N. Engl. J. Med.* **389**, 2221–2232 (2023).
85. Q. Huang, H. Tang, K. Wang, R. Li, C. Eldeniz, N. Nguyen, T. H. Schindler, L. R. Peterson, Y. Yang, Y. Yan, J. Cheng, P. K. Woodard, J. Zheng, Model-based self-supervised learning for quantitative assessment of myocardial oxygen extraction fraction and myocardial blood volume. *Magn. Reson. Med.* **94**, 1793–1803 (2025).
86. J. Cho, J. Zhang, P. Spincemaille, H. Zhang, S. Hubertus, Y. Wen, R. Jafari, S. Zhang, T. D. Nguyen, A. V. Dimov, A. Gupta, Y. Wang, QQ-NET—Using deep learning to solve quantitative susceptibility mapping and quantitative blood oxygen level dependent magnitude (QSM+qBOLD or QQ) based oxygen extraction fraction (OEF) mapping. *Magn. Reson. Med.* **87**, 1583–1594 (2022).
87. X. He, D. A. Yablonskiy, Quantitative BOLD: Mapping of human cerebral deoxygenated blood volume and oxygen extraction fraction: Default state. *Magn. Reson. Med.* **57**, 115–126 (2007).
88. K. Thygesen, J. S. Alpert, A. S. Jaffe, B. R. Chaitman, J. J. Bax, D. A. Morrow, H. D. White, Executive Group on behalf of the Joint European Society of Cardiology (ESC)/American College of Cardiology (ACC)/American Heart Association (AHA)/World Heart Federation (WHF) Task Force for the Universal Definition of Myocardial Infarction, Fourth universal definition of myocardial infarction (2018). *Circulation* **138**, e618–e651 (2018).
89. W. Ganz, H. J. Swan, Measurement of blood flow by thermodilution. *Am. J. Cardiol.* **29**, 241–246 (1972).
90. G. K. Lund, M. F. Wendland, A. Shimakawa, H. Arheden, F. Ståhlberg, C. B. Higgins, M. Saeed, Coronary sinus flow measurement by means of velocity-encoded cine MR imaging: Validation by using flow probes in dogs. *Radiology* **217**, 487–493 (2000).
91. J. He, Q. Liu, A. G. Christodoulou, C. Ma, F. Lam, Z. P. Liang, Accelerated high-dimensional MR imaging with sparse sampling using low-rank tensors. *IEEE Trans. Med. Imaging* **35**, 2119–2129 (2016).
92. Y. Huang, X. Guan, X. Zhang, G. Yoosefian, H. Ho, L. T. Huang, H. Y. Lin, G. Anthony, H. L. Lee, X. Bi, F. Han, S. F. Chan, K. P. Vora, B. Sharif, D. P. Singh, K. Youssef, D. Li, H. Han, A. G. Christodoulou, R. Dharmakumar, H. J. Yang, Accurate intramyocardial hemorrhage assessment with fast, free-running, cardiac quantitative susceptibility mapping. *Radiol. Cardiothorac. Imaging* **6**, e230376 (2024).
93. X. Guan, H. J. Yang, X. Zhang, N. Wang, H. Han, R. Tang, Z. Hu, K. Youssef, K. Vora, M. S. Krishnam, A. G. Christodoulou, D. Li, B. Sharif, R. Dharmakumar, Non-electrocardiogram-gated, free-breathing, off-resonance reduced, high-resolution, whole-heart myocardial T(2) * mapping at 3 T within 5 min. *Magn. Reson. Med.* **91**, 1936–1950 (2024).
94. S. Portnoy, N. Milligan, M. Seed, J. G. Sled, C. K. Macgowan, Human umbilical cord blood relaxation times and susceptibility at 3 T. *Magn. Reson. Med.* **79**, 3194–3206 (2018).
95. A. Suinesiaputra, D. A. Bluemke, B. R. Cowan, M. G. Friedrich, C. M. Kramer, R. Kwong, S. Plein, J. Schulz-Menger, J. J. M. Westenberg, A. A. Young, E. Nagel, Quantification of LV function and mass by cardiovascular magnetic resonance: Multi-center variability and consensus contours. *J. Cardiovasc. Magn. Reson.* **17**, 63 (2015).
96. P. M. Heerdt, S. Korfhagen, H. Ezz, C. Oromendia, Accuracy of a simulation algorithm for modelling LV contractility, diastolic capacitance, and energetics using data available from common hemodynamic monitors and echocardiography. *J. Cardiothorac. Vasc. Anesth.* **32**, 381–388 (2018).
97. D. Stevenson, J. Revie, J. G. Chase, C. E. Hann, G. M. Shaw, B. Lambermont, A. Ghuyen, P. Kolh, T. Desai, Beat-to-beat estimation of the continuous left and right cardiac elastance from metrics commonly available in clinical settings. *Biomed. Eng. Online* **11**, 73 (2012).

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Quantifying myocardial oxygen consumption and efficiency with motion-resolved cardiac MRI

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Editor's summary

How the heart uses oxygen is important to understand for many pathologies. Huang *et al.* developed a noninvasive magnetic resonance imaging (MRI) technique to quantify myocardial oxygen consumption to address complex calibration and motion artifacts that often make clinical MRI difficult to use. The MRI technique is self-calibrating, high resolution, and motion resolved, which allowed quantification of myocardial oxygen extraction from the blood within 3 min. Simulations and validation in a porcine model were carried out to develop the MRI technique. Impaired oxygen consumption was then observed in patients with heart failure phenotypes compared with controls, and accuracy was validated in three individuals who underwent MRI in addition to cardiac blood sample collection for oxygen saturation testing. Together, these data show how an MRI technique could be used to determine cardiac oxygen consumption with ease and speed in the clinic. —Brandon Berry

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